IEE-017

DEVCOM-Soldier Center

Audio Noise Inverter

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Project Requirements

Project Overview

The goal of this project is to manage an electrical engineering team tasked with designing and developing a working audio noise inverter prototype capable of taking an audio input and producing an output that effectively cancels out the users' sound from their environment. The capstone teams will work in conjunction with DEVCOM-Soldier Center throughout the course of this project.

Project Deliverables

- Create a **Scope Statement** to define team goals and clarify project expectations
- Develop and maintain **Project Logs** to monitor and control project schedule, budget, and scope
- Define project deadlines and identify the critical path using a **Project Timeline**
- Use a **Risk Assessment** to identify, assess, and mitigate risks to the project
- Provide a central source for main project management documents in a **Project Plan**

Design Description

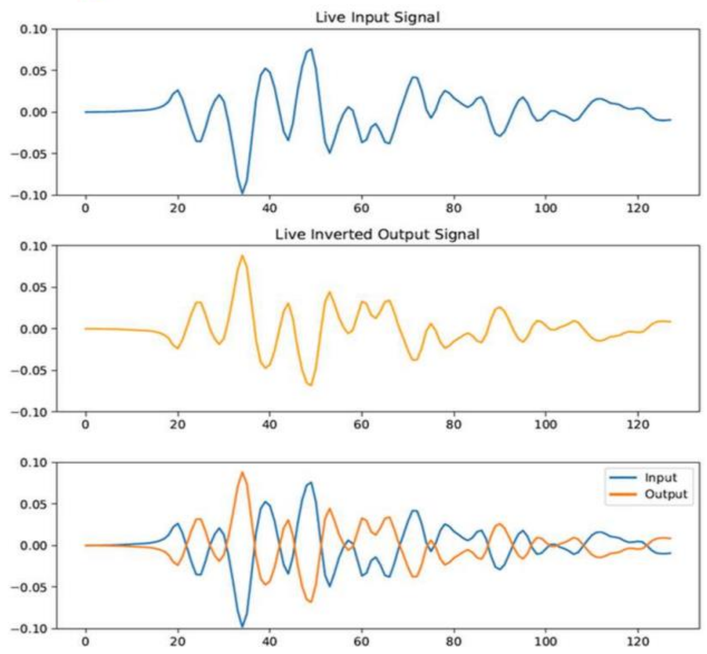
A **schedule control chart**, tracks project tasks and shows the **difference between expected and actual dates**. This chart allows the team to monitor team performance over time and quickly flag possible delays, allowing for effective mitigation. A **risk assessment** was used to identify potential risks and develop mitigation plans. This approach is justified because it allows the management team to **determine which risks have the highest priority**, so they can be addressed first.

Results

Schedule Control Chart

Task	Expected End Date	Actual End Date
Bill of Materials	10/13/2025	11/11/2025
Proof of Concept	12/1/2025	12/2/2025
Software Development	2/13/2026	2/10/2026
Software Testing & Modifying	3/1/2026	2/27/2026
Hardware Development	2/23/2026	2/26/2026
Hardware Testing & Modifying	3/6/2026	3/9/2026
Integration of Software & hardware	3/7/2026	3/11/2026
Final Prototype Testing & Modifying	4/21/2026	4/8/2026

Result: 13 days ahead of schedule



FMEA Risk Assessment

Risk	Description	RPN	Effect	Mitigation Plan	New RPN
Software Deployment	Software program does not function as intended when deployed onto hardware	140	Non-functional prototype **(scope)**	Preliminary research, verification, and testing of software limitations, capabilities, and compatibility with hardware.	42
Proof of Concept	Poor project design	126	Project redesign & schedule delay **(scope & schedule)**	Additional research & further verification of design functionality and capability.	18
Hardware	Components break during testing	21	Hardware testing delayed **(schedule & budget)**	Order backup components	6
Bill of Materials	Missing components	20	Hardware construction delayed **(budget & schedule)**	BOM seeing several rounds of verification before official review & submission	5

Result: 77% decrease in risk exposure

Budget Control

Part	Qty	US $
Raspberry Pi5 Microcontroller + Cooling Fan	1	$90.50
Raspberry Pi Steel Case	1	$29.90
Audio Converter	1	$149.80
32GB storage	1	$12.68
Microphone	1	$22.20
Amplifier	1	$54.93
Power Bank	1	$429.98
Power Adapter	1	$34.95
Full Range Speaker	1	$55.73
	Total:	$805.77

Result: $194.23 underbudget

Lessons Learned

- Managing online cross-functional teams
- Developing achievable goals around constraints
- Asking for and implementing feedback

Future Work

The prototype will be further developed by researchers at DEVCOM soldier center.

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